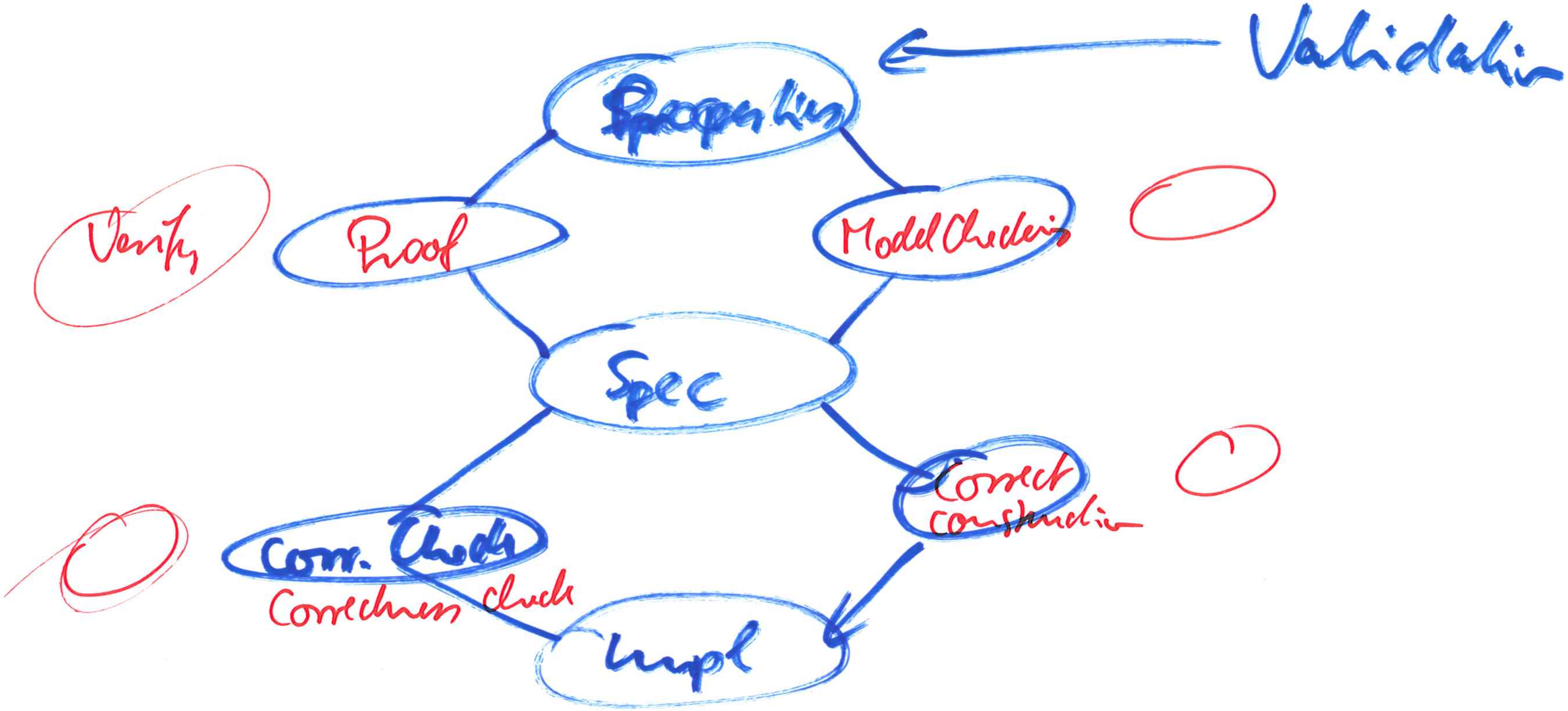
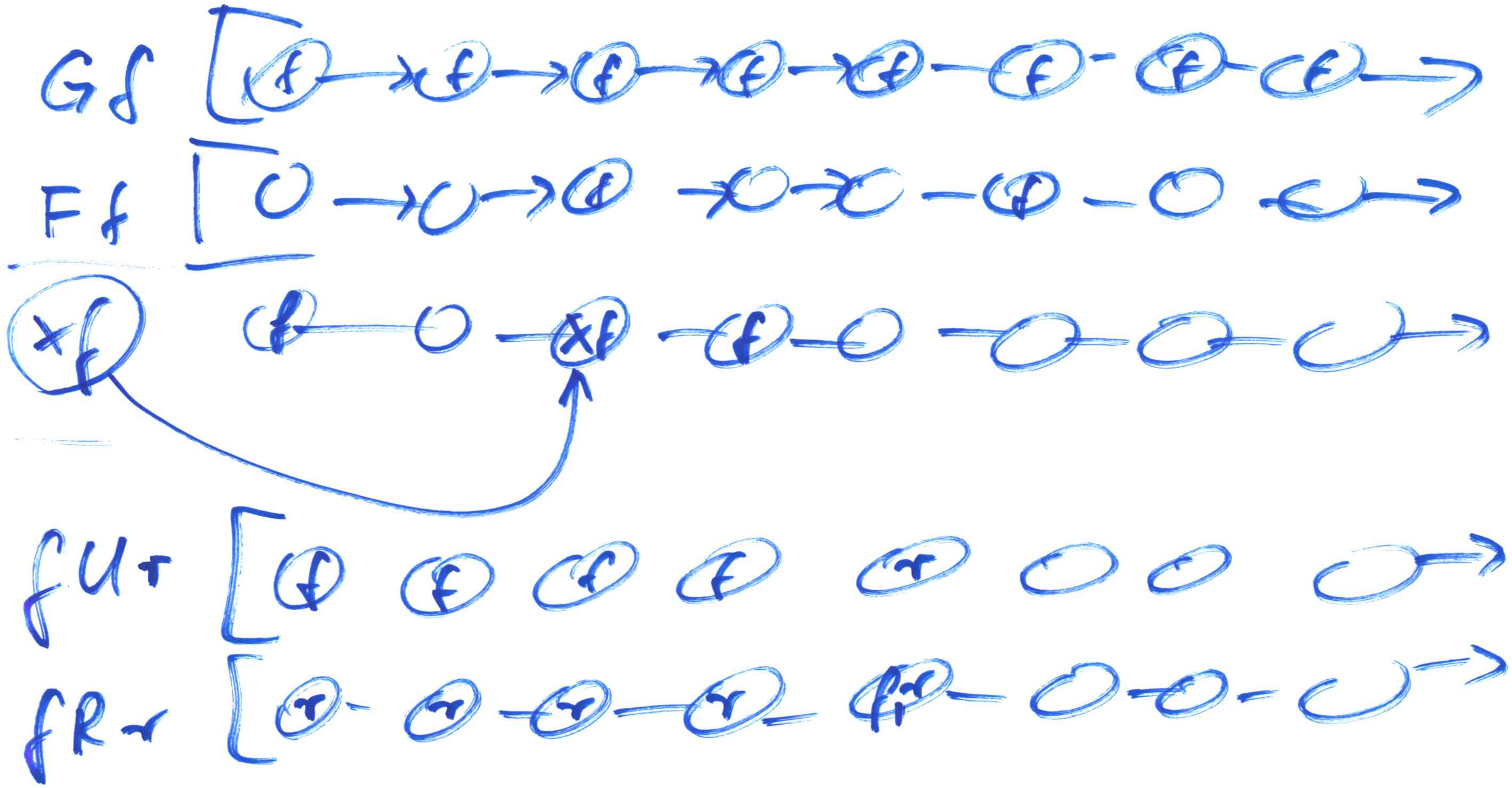

Validation
Properties
Verify
Proof
Model Checking
Spec
Correct construction
Corr. Check
Correctness check
Impl


Binary
Validation

$G\,(x=5)$ $\qquad$ $G\,X\,(x=5)$



```
public interface Peano {
    static Peano zero();
    static Peano succ(Peano p);
    static Peano pred(Peano p);
    static Peano add(Peano p, q);
}

public class PeanoImpl implements Peano ...


Peano p = new PeanoImpl();
PeanoImpl pi = new PeanoImplBest();
```

{Pre} Code {Post}

$$x, y := x', y' \mid x' = y \wedge y' = x \quad \sqsubseteq$$

$$\mathrm{VAR}\ t : T \bullet t := x\ ;\ x, y, t := x', y', t' \mid x' = y \wedge y' = x$$

$Max(x, y, z : NUM) \quad z := z' \mid z' = \max x\, y$

---

if $(x > y)$ then $\{x > y\}$ $z := z' \mid z' = \max x\, y$

else $\{x <= y\}$ $z := z' \mid z' = \max x\, y$

$x > y \Rightarrow \max x\, y = x$

---

if $(x > y)$ $\{\max x\, y = x\}$ $z := z' \mid z' = \max x\, y$

else $\{\max x\, y = y\}$ $z := z' \mid z' = \max x\, y$

---

if $(x > y)$ $z := z' \mid z' = x$

else $z := z' \mid z' = y$

---

if $(x > y)$ $z := x$

else $z := y$